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Bayesian Inference

Previous Missions Report

Introduction

The following notes summarize a small number of previous search missions carried out in environments considered broadly similar to the one studied in this deliverable. These reports are **qualitative** and **incomplete**: they are not part of the formal dataset and should not be treated as exact calibration data.

Their purpose is only to provide contextual information that may help you reason about how detectability could depend on mission effort and environmental conditions such as depth and seafloor roughness.

Mission report 1

A relatively large area was inspected with a **low effort level**. Most of the searched cells were located in a **moderately deep zone** with **smooth or only mildly irregular terrain**. No object was detected during the mission.

The team responsible for the search noted that the covered area was wide, but that the inspection intensity was limited. Their conclusion was that the negative outcome could not be interpreted as strong evidence that the object was absent from the searched region.

Mission report 2

A smaller and more focused area was inspected with a **high effort level**. The region was described as **shallow to moderately deep**, with **low roughness** and relatively favorable inspection conditions. The object of interest was detected quickly after the beginning of the mission.

The mission log emphasized that the combination of concentrated coverage, favorable terrain, and high effort appeared to produce a high detection capability.

Mission report 3

A mission with **high effort** was carried out over a region known to be **deep** and **highly irregular**. Despite the intensity of the search, no object was detected.

According to the final note of the report, the negative result was not considered surprising. The team highlighted that strong depth effects and rugged terrain may severely reduce the practical efficiency of the search, even when effort is high.

Mission report 4

A medium-sized region was inspected with **intermediate effort**. The environment was described as **moderately shallow**, but with **noticeable seafloor irregularities**. The search produced an uncertain outcome at first, and only after repeated inspection of the same area was a positive detection obtained.

The report suggested that, under intermediate conditions, detectability may not be negligible, but can still be substantially lower than in smooth and shallow regions.

General remark

Taken together, these reports suggest that mission effort, depth, and roughness may all play a role in detectability. However, the reports do not determine a unique mathematical model, nor do they imply that every relevant variable must be used. Any model proposed in the deliverable should therefore be justified on probabilistic and practical grounds.